Two-Sample t-Test Assuming Equal Variances

Complete Line-by-Line Calculation

No.	Group A	Group B
1	72	80
2	75	82
3	78	85
4	70	79
5	74	84
6	77	81
7	73	83
8	76	86
9	71	80
10	75	82
11	79	84
12	74	81

1. Number of Observations

Group A: n■ = 12
Group B: n■ = 12

2. Mean of Group A

Formula: X■■ = ΣX■ / n■
Sum = 72 + 75 + 78 + 70 + 74 + 77 + 73 + 76 + 71 + 75 + 79 + 74 = 894
X■■ = 894 / 12 = **74.5**

3. Mean of Group B

Formula: X■■ = ΣX■ / n■
Sum = 80 + 82 + 85 + 79 + 84 + 81 + 83 + 86 + 80 + 82 + 84 + 81 = 987
X■■ = 987 / 12 = **82.25**

4. Difference Between Means

X■■ – X■■ = 74.5 – 82.25 = –**7.75**
Group A is 7.75 points lower than Group B.

5. Variance of Group A

Formula: s■^{2} = Σ(X■ – X■■)^2 / (n■ – 1)
X■■ = 74.5
Squared deviations:
6.25 + 0.25 + 12.25 + 20.25 + 0.25 + 6.25 + 2.25 + 2.25 + 12.25 + 0.25 + 20.25 + 0.25 = 83

$s_{\blacksquare}^2 = 83 / 11 =$ **7.54545**

6. Variance of Group B

Formula: $s_{\blacksquare}^2 = \Sigma(X_{\blacksquare} – X_{\blacksquare\blacksquare})^2 / (n_{\blacksquare} – 1)$
$X_{\blacksquare\blacksquare} = 82.25$
Sum of squared deviations = 52.25
$s_{\blacksquare}^2 = 52.25 / 11 =$ **4.75**

7. Pooled Variance

Because this is an equal-variance test:
$S_{\blacksquare}^2 = [(n_{\blacksquare} – 1)s_{\blacksquare}^2 + (n_{\blacksquare} – 1)s_{\blacksquare}^2] / (n_{\blacksquare} + n_{\blacksquare} – 2)$
$S_{\blacksquare}^2 = [(11)(7.54545) + (11)(4.75)] / (12 + 12 – 2)$
$S_{\blacksquare}^2 = (83 + 52.25) / 22$
$S_{\blacksquare}^2 = 135.25 / 22 =$ **6.147727**

8. Hypothesized Mean Difference

Excel uses **0** because the null hypothesis assumes equal population means:
$H_{\blacksquare}: \mu_{\blacksquare} – \mu_{\blacksquare} = 0$

9. Degrees of Freedom

Formula: $df = n_{\blacksquare} + n_{\blacksquare} – 2$
$df = 12 + 12 – 2 =$ **22**

10. Standard Error

Formula: $SE = \sqrt{[S_{\blacksquare}^2(1/n_{\blacksquare} + 1/n_{\blacksquare})]}$
$SE = \sqrt{[6.147727(1/12 + 1/12)]}$
$1/12 + 1/12 = 0.166667$
$SE = \sqrt{(6.147727 \times 0.166667)}$
$SE = \sqrt{1.024621} \approx$ **1.01223**

11. t-Statistic

Formula: $t = [(X_{\blacksquare\blacksquare} – X_{\blacksquare\blacksquare}) – 0] / SE$
$t = [(74.5 – 82.25) – 0] / 1.01223$
$t = –7.75 / 1.01223$
$t =$ **–7.6563**
This matches Excel: t Stat = –7.656319198.

12. Why is t Negative?

Because Group A mean is smaller than Group B mean:
74.5 – 82.25 = –7.75
For a two-tailed test, use the absolute value: $|t| =$ **7.6563**.

13. One-Tailed p-Value

Excel gives:
6.0705E–08
This is approximately 0.000000060705.

14. Two-Tailed p-Value

Excel gives:
1.2141E–07
This is approximately 0.00000012141.

15. Critical t-Value

With $\alpha = 0.05$ and df = 22:
One-tail critical t = **1.717144374**
Two-tail critical t = **2.073873068**

16. Final Decision Using t

|t Stat| = 7.6563
Two-tail critical t = 2.0739
7.6563 > 2.0739
Therefore, reject H■.

17. Final Decision Using p-Value

Two-tail p-value = 1.2141E–07
$\alpha = 0.05$
1.2141E–07 < 0.05
Therefore, reject H■.

18. Final Conclusion

There is a statistically significant difference between Group A and Group B.

Group A mean = 74.5
Group B mean = 82.25
Difference = 7.75 points

Group B has a significantly higher mean than Group A at the 5% significance level.

Quick Formula Summary

Step	Formula	Your Result
1	X■■ = ΣX■ / n■	74.5
2	X■■ = ΣX■ / n■	82.25
3	s■² = Σ(X■–X■■)²/(n■–1)	7.54545
4	s■² = Σ(X■–X■■)²/(n■–1)	4.75
5	Pooled Variance	6.147727
6	SE = √[S■²(1/n■+1/n■)]	≈1.01223
7	t = (X■■–X■■)/SE	–7.6563
8	df = n■+n■–2	22
9	Two-tail p-value	1.2141E–07

10	Two-tail critical t	2.073873
11	Decision	Reject H■